AY2026 Semester 1

C245

Data Analytics with Gen AI

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Content

- RP’s Graduate Profile
- Module Information
- Module Outcomes
- Module Summary
- Attendance
- Assessment
- AI Usage

RP's Graduate Profile

Resilient Problem Solver

ARTICULATE SELF-STARTER

INITIATE ACTION

- RP Graduates are **skilled communicators** who excel in their ventures/pursuits. Their clarity, conviction and enthusiasm drive them to **develop unique solutions** to navigate challenges.

PASSIONATE COMMUNITY BUILDER

IMPROVE LIVES

- RP Graduates are committed to creating positive impact in local and global communities. They **care deeply** and contribute generously to make a difference in the lives of others.

PURPOSEFUL GAME-CHANGER

INSPIRE INNOVATION

- RP Graduates are curious individuals with the creativity to solve problems and seize opportunities. Their **enterprising spirit and resourcefulness** help to transform ideas into reality.

RP's Graduate Profile

ARTICULATE SELF-STARTER
- Skilled **communicator**
- Develops **unique solutions**

PASSIONATE COMMUNITY BUILDER
- **Cares deeply**
- Creates **positive impact**

PURPOSEFUL GAME-CHANGER
- **Enterprising**
- Transforms **ideas into reality**

The 6 Graduate Profile competencies

RP aims to develop all students into **resilient problem solvers** who are **articulate self-starters, passionate community builders**, and **purposeful game-changers**. Throughout your studies, you will have ample opportunities to acquire and strengthen RP's Graduate Profile competencies.

Module Information

Module Objectives

This module introduces students to the capabilities of Data Analytics and Generative AI (GenAI), equipping them with foundational skills to uncover insights, create impactful data visualizations, and support informed business decision-making. Students will explore how GenAI enhances data analysis, generates meaningful insights, and strengthens strategic decision-making across organizational contexts. Through practical, hands-on activities, they will develop competencies in data storytelling, dashboard creation, and visualization design, learning to communicate complex data clearly and effectively. By the end of this module, students will be able to apply AI-driven analytics to address real-world business challenges, preparing them to thrive in a data-driven future.

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Module Outcome and Learning Outcome (MO/LO)

Module Outcome (MO)	Learning Objective (LO)
1. Foundation of Data Analytics Identify the importance of data-driven decisions to an organization and determine the impact of data analytics to the organization strategy	• Describe and analyze the key components and architecture of a Data Analytics environment. • Explain and apply the phases of a Data Analytics project (requirements, measures/dimensions, analysis, reporting, maintenance). • Classify data types (ordinal, nominal, interval, ratio) and apply them to appropriate analytical contexts. • Demonstrate how GenAI can assist in requirements gathering, querying, analysis, and report generation through prompt engineering. • Apply prompt engineering (COSTAR/RISEN) and storytelling structure (DIVA) to communicate data insights effectively
2. Generate graphs based on business objectives. e.g Bar, Pie, Line Based on the visual charts, interpret and give insights that are help in making business decisions	• Compare pros and cons of different chart types (bar, pie, line, scatter, boxplot, heatmap). • Translate business requirements into questions and design dashboards that answer them. • Explain the importance of interactivity and storytelling in management reporting. • Create and interpret visualizations that address business objectives, and communicate data-driven insights to support decision-making using appropriate chart types and GenAI tools.

Module Outcome and Learning Outcome (MO/LO)

Module Outcome (MO)	Learning Objective (LO)
3. Apply statistical and GenAI tools to analyze data, uncover patterns, and generate insights that support informed decision-making. Students will use applications such as Excel and SPSS/Jamovi/JASP, supported by Generative AI, to perform descriptive, relational, and inferential analyses. The focus is on interpreting results and deriving meaningful business insights.	• Explore and visualize data distributions using frequency tables, crosstabs, and AI-assisted visual summaries to identify key patterns and anomalies. • Summarize and interpret data using descriptive statistics (central tendency, variability, boxplots) and apply GenAI tools to generate narrative summaries of results. • Analyze and interpret relationships between variables using scatterplots, correlation, and regression; leverage GenAI to explain patterns and predictive trends. • Apply and interpret probability distributions (binomial, Poisson, normal) using GenAI-assisted simulation and contextual explanation in business scenarios. • Perform and interpret inferential tests (t-test, ANOVA, chi-square) using Excel/SPSS outputs, and use GenAI to translate results into plain-language insights. • Construct and evaluate confidence intervals, interpreting sampling accuracy and variability with GenAI-generated interpretations and recommendations. • Integrate GenAI across all statistical stages to simulate outcomes, validate results, and generate clear, data-driven recommendations for decision-making.
4. Examine how Agentic AI transforms traditional data analytics into an intelligent, goal-driven process that enhances decision-making and human-AI collaboration.	• Explain what Agentic AI is and how it differs from traditional analytics and generative AI. • Describe the components of an Agentic AI system (e.g., planning, reasoning, action, feedback). • Compare the traditional analytics workflow with an Agentic AI-driven workflow. • Reflect on how Agentic AI can augment — not replace — human decision-making in organizations.

Lesson Structure

CA	Week	Lesson	AY2026
CA1	1	1	Introduction to Data Analytics, NOIR
		2	Understanding Business Concerns, Business Questions, Analytical Questions and Observations
	2	3	Visualization I
		4	Visualization 2
	3	5	Visualization 3
		6	PS
	4	7 (eLearning)	COSTAR/RISEN
		8(eLearning)	DIVA
	5	9	Central Tendancies I
		10	Central Tendancies II
	6	11	Linear Regression
		12 (CA1)	Probability + CA1
	7	13	Binomial
		14	Poisson

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Lesson Structure

CA	Week	Lesson	AY2026
CA 2	8	15	Normal Distribution
		16	PS
	9	17	CI
		18	HT
	10	19	T-Test
		20	T-Test
	11	21	ANOVA
		22	CHI-Square
	12	23	Wrap PS
		24 (CA2)	Wrap PS + CA2
	13	25 (eLearning) (CS)	Agentic AI in Data Analytics
		26 (eLearning) (CS)	Agentic AI with n8n in Data Analytics

- Lesson presentation slides will be made available during the lesson.
- All deliverables must be submitted by Sunday, 23:59, within the same week of the lesson.
- Solution slides will be released on the following Monday for selected problem statements only.

Assessment Guidelines

Allowed:

- Excel Helper provided for the module
- Lecture notes and lesson materials

Not Allowed:

- Docker image
- Generative AI tools (e.g. ChatGPT, Copilot, etc.)
- External code repositories or pre-built solutions
- Communication with others during the assessment

Note:

- Only tools and resources explicitly provided and permitted may be used.
- The Docker image is **not allowed** for use during assessments.
- Use of any unauthorized tools may constitute an academic integrity violation.

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Attendance

- Arriving after 15 minutes grace period or missed more than 15 minutes of the lesson without lecturer’s permission will be marked as “Partial”.
- Reminder that attendance will be taken for all ***scheduled* synchronous (face-to-face) lessons <u>and</u> *scheduled* asynchronous e-learning lessons**; attendance will <u>not</u> be taken for <u>unscheduled</u> asynchronous e-learning lessons.
- Even when you have an LOA, you must submit deliverables in eLearning, or you will be marked as absent.
- Unscheduled asynchronous e-learning lessons are integral to the curriculum and will be assessed in Continuous Assessments and/or Final Assessment
- Attendance policy summary, implications on
  - Total Absences without LOA more than 20%
  - Total Absences > 50%
- There will no longer be debarment from exams. For students who do not meet the new attendance requirements, the consequences are:
  - they must pass Final Assessment;
  - their module grade may be capped; and
  - in extreme cases of poor attendance, they may fail the module.

Refer to the relevant section in the student handbook here: [Student Handbook - Absenteeism (sharepoint.com)]

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What Happened If I Am Absent?

- Week 6 Lesson 12 – Quiz (CA1)
- Week 12 Lesson 24 – Quiz (CA2)
- Week 13 Lesson 25 and 26 – Interview (Communication Skill)
- If you are absent with an approved Letter of Absence (LOA), your make-up CA (written and interview components) will be scheduled later.
- Failure to turn up for the make-up CA or CS will result in 0 mark for CA and will impact your module grade.

Assessment/ CA Guidelines

Component		Weighting (%)
CA1	Online Quiz (KS)	10%
CA2	Online Quiz(KS)	10%
	Presentation(CS)	10%
SDLC	Self-Directed and Collaborative Learning	15%
FA	Final Assessment	55%
	Total = CA1 + CA2 + SDCL + FA	100%

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Communication Skills (CS) Rubrics

PERFORMANCE CRITERIA	DESCRIPTION	PERFORMANCE LEVELS			
		Excellent (4 marks)	Good (3 marks)	Fair (2 marks)	Unsatisfactory (0 – 1 mark)
COMMUNICATION SKILLS (CS) *The learner **conveys** information and ideas associated with KS to an audience, in a **clear, confident and engaging** manner.*	<u>Engagement</u> • Builds on key points or messages that result in a **well-organised narrative** to meet the **planned communication objective(s)**.	<u>Engagement</u> • Narrative builds on **all** key points or messages that are **well-organised in all parts, meeting** the planned communication objective(s).	<u>Engagement</u> • Narrative builds on **most** key points or messages that are **well-organised in most parts, reasonably meeting** the planned communication objective(s).	<u>Engagement</u> • Narrative builds on **some** key points or messages that are **organised in some parts, nominally meeting** the planned communication objective(s).	<u>Engagement</u> • Narrative **does not** build on key points or messages, or they are **disorganised in most or all parts, and does not meet the planned communication objective(s).**
	• **Maintains the audience's interest** throughout the presentation by using a **variety of techniques** such as storytelling, humor, anecdotes, or real-life examples.	• Uses **appropriate techniques** to sustain audience interest **throughout** the presentation.	• Uses **appropriate techniques** to sustain audience interest **for most part** of the presentation.	• Uses **appropriate techniques** to sustain audience interest **for some parts** of the presentation.	• **Does not use any technique** to sustain audience interest throughout the presentation.
	<u>Clarity</u> • Exhibits **clarity in diction**, with appropriate **pace** (rate of speech), **volume** (loudness or softness of voice), **pitch** ("high" or "low" tone of voice) and **pauses** (in between words or sentences).	<u>Clarity</u> • Exhibits **very clear diction** with **excellent** pace, volume, pitch and pauses **throughout** the presentation.	<u>Clarity</u> • Exhibits **clear diction** with **good** pace, volume, pitch and pauses **for most part** of the presentation.	<u>Clarity</u> • Exhibits **clear diction, albeit** with **inappropriate** pace, volume, pitch and pauses **for some parts** of the presentation.	<u>Clarity</u> • Exhibits **unclear diction** with **inappropriate** pace, volume, pitch and pauses for **most or all parts** of the presentation.
	• Demonstrates **smooth delivery** without losing train of thought, stuttering, or unnecessarily repeating points.	• **Delivers presentation smoothly without** losing train of thought, stuttering, or unnecessarily repeating points **throughout** the presentation.	• **Delivers presentation smoothly without** losing train of thought, stuttering, or unnecessarily repeating points **for most part** of the presentation.	• **Delivers presentation**, losing train of thought, stuttering, or unnecessarily repeating points **for some parts** of the presentation.	• **Delivers presentation stiffly**, losing train of thought, stuttering, or unnecessarily repeating points **throughout** the presentation.
	• Uses **domain-specific/technical terms or references** etc. appropriately, contributing to the **clarity and impact** of communication.	• Uses domain-specific/technical terms or references etc. appropriately **throughout**, resulting in a **highly impactful** communication.	• Uses domain-specific/technical terms or references etc. appropriately **most of the time**, resulting in **an impactful** communication.	• Uses domain-specific/technical terms or references etc. appropriately **sometimes**, resulting in **communication with some impact.**	• **Does not use** domain-specific/technical terms or references etc., resulting in communication with **no impact**.
	<u>Confidence</u> • Projects confidence through **eye contact** (holding the audience's gaze), without looking at notes.	<u>Confidence</u> • **Maintains eye contact** with audience **throughout** and **does not** read from notes.	<u>Confidence</u> • **Maintains eye contact** with audience **most of the time, seldom** reading from notes.	<u>Confidence</u> • **Maintains eye contact** with audience **sometimes**, **mostly** reading from notes.	<u>Confidence</u> • **Maintains no eye contact** with audience, **entirely** reading from notes.
	• Displays **excellent body language** (upright posture, natural gestures and avoiding bad habits such as touching hair etc.)	• Displays **excellent body language**.	• Displays **good** body language.	• Displays **mostly appropriate** body language.	• Displays **inappropriate** body language.

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SDCL Rubric

PERFORMANCE CRITERIA	DESCRIPTION	PERFORMANCE LEVELS			
		Excellent (4 marks)	**Good (3 marks)**	**Fair (2 marks)**	**Unsatisfactory (0 – 1 mark)**
SELF-DIRECTED COLLABORATIVE LEARNING (SDCL) *The learner **engages** with the **learning processes**, in order to effectively **support** the attainment of **KS**.*	**SELF-DIRECTED LEARNING (SD)** 50% of SDCL score **Self-Managing** • Sets and achieves relevant goals and related learning outcomes. • Applies feedback given to improve learning.	**Self-Managing** • **Sets and achieves** relevant goals and related learning outcomes **independently**. • **Takes charge** of learning tasks (pro-active), e.g., deliverables and assignments, and demonstrates **clarity in approach throughout**, e.g., being mindful of established timelines and expectations, seeking clarifications when unsure etc., **independently**.	**Self-Managing** • **Sets and achieves** relevant goals and related learning outcomes, with **guidance**. • **Mostly takes charge** of learning tasks (pro-active), e.g., deliverables and assignments, and demonstrates **clarity in approach in most parts**, e.g., being mindful of established timelines and expectations, seeking clarifications when unsure etc., **with guidance**.	**Self-Managing** • Sets goals, of which **most are relevant**, and **attempts to achieve** goals and related learning outcomes, with **close supervision**. • **Occasionally takes charge** of learning tasks (pro-active), e.g., deliverables and assignments, and demonstrates **clarity in approach in some parts**, e.g., being mindful of established timelines and expectations, seeking clarifications when unsure etc., **with close supervision**.	**Self-Managing** • **Does not** set goals and/or attempt to achieve goals and related learning outcomes, even with **close supervision**. • **Does not take charge** of learning tasks (pro-active), e.g., deliverables and assignments, and **does not demonstrate clarity in approach**, e.g., not being mindful of established timelines and expectations, and not seeking clarifications when unsure etc., **even with close supervision**.
	Self-Monitoring • Reflects on learning processes and progress. • Adjusts learning strategies and approaches to enable improvement in learning outcomes.	**Self-Monitoring** • Reflects on learning processes and progress, with understanding of one's strengths, weaknesses, and areas for improvement **independently.** • Adjusts learning strategies and approaches based on reflection and/or feedback given, resulting in **attainment of most learning outcomes, or excellent improvement in learning.**	**Self-Monitoring** • Reflects on learning processes and progress, with understanding of one's strengths or weaknesses, and areas for improvement, **with guidance.** • Adjusts learning strategies and approaches based on reflection and/or feedback given, resulting in **attainment of adequate learning outcomes, or good improvement in learning.**	**Self-Monitoring** • Reflects on learning processes and progress, with **some understanding** of one's strengths or weaknesses, **with close supervision**. • Adjusts learning strategies and approaches based on reflection and/or feedback given, resulting in **attainment of baseline learning outcomes, or reasonable improvement in learning.**	**Self-Monitoring** • **Does not** reflect on one's learning processes and progress, **even with close supervision**. • **Does not** adjust learning strategies and approaches, when necessary, to support attainment of learning outcomes or improvement in learning.
	COLLABORATIVE LEARNING (CL) 50% of SDCL score • Leads or works with the team to achieve their collective goal(s). • Completes the assigned tasks/roles. • Contributes prior knowledge, experiences, learning resources or asks relevant questions to support the rigour of knowledge construction/ skill development during lessons.	• **Facilitates discussions** to enable team to **establish** unified goal(s). • Completes assigned tasks or responsibilities **throughout** the learning process, and **often takes on additional responsibilities**. • Contributes **highly relevant** (subject matter alignment) **and useful** (applicability to context) prior knowledge, experiences, resources, and/or asks **highly relevant** questions to support team learning.	• **Contributes to discussions** to enable team to establish unified goal(s). • Completes assigned tasks or responsibilities **throughout** the learning process. • Contributes **relevant** (subject matter alignment) **and useful** (applicability to context) prior knowledge, experiences, resources, and/or asks **relevant** questions to support team learning.	• **Participates minimally** in the establishment of unified goal(s) and **displays awareness** of these collective goal(s). • Completes assigned tasks or responsibilities in **most parts** of the learning process. • Contributes **some relevant** (subject matter alignment) prior knowledge, experiences, resources, and/or asks **some relevant** questions to support team learning.	• **Does not** participate in the establishment of unified goal(s), and/or does not display awareness of these goal(s). • **Does not** complete assigned tasks or responsibilities throughout, or for all/most part of the learning process. • **Does not** contribute relevant (subject matter alignment) or useful (applicability to context) prior knowledge, experiences, resources, and/or **does not** ask relevant questions to support team learning.

Plagiarism & Academic Dishonesty

Warning:

Any student caught infringing any of the Assessment Rules and Regulations in any way is liable to disciplinary action including, but not limited to, an outright failure for the Assessment and downgrading of the Module Grade. More severe penalties which can be imposed on a student caught cheating in a summative assessment include suspension or even dismissal from the Polytechnic.

Student Handbook - Summative_assessment_rules_and_regulations (sharepoint.com)

IT Tools

POLITEMall:

- Lesson Materials (Lesson Slides, Problem Statement, Lab, WS)
- Solution Slides (Released after lesson week)

Software Tools:

- Tableau
- Docker Desktop + Image
- JASP

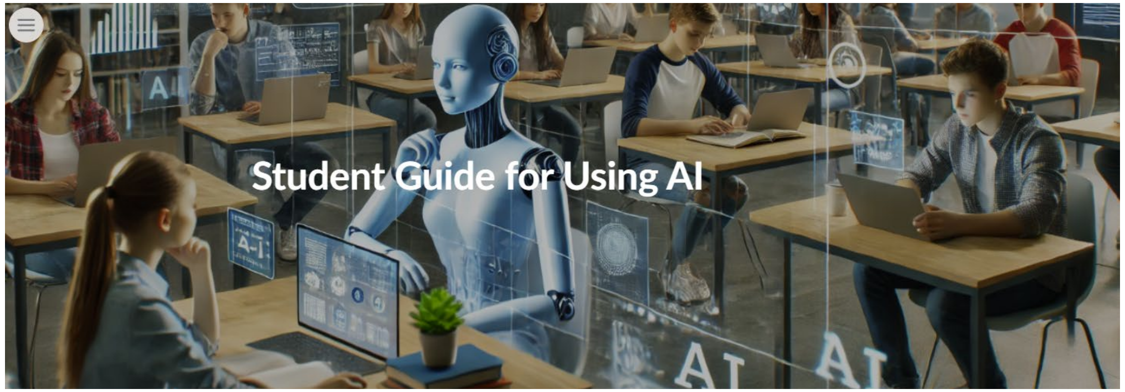
EdTech Tools:

- MS Team
- Lesson Explainer Video
- Excel
- LinkedIn Learning, YouTube Videos, Kahoot!
- SCORM Package + Video (eLearning, PS Solutions)

IT Issues (for students)

- Students to email help-IT@rp.edu.sg, RP-connect App or call 66971500.
- Helpdesk counter in the library

AI Usage



[Click here] to read the Student Guide for Using AI

[Click here] to read RP's assessment rules & regulations

At RP, you are encouraged to use AI tools for learning, but you are responsible for using them ethically. This means maintaining academic integrity by submitting original work that reflects your understanding. It also requires critically evaluating AI-generated content to avoid misleading or biased outputs.

Use AI wisely to enhance your learning, not replace your thinking.

Students are encouraged to use Generative AI to support their learning; however, it must not be used in assessments or classwork submissions. For classwork, students are expected to submit their own thoughts and analysis.

Learning assistance

- For this module, an Excel Helper and Docker image are provided to support your statistical work.
- As with any application, issues may arise and will be addressed as they are identified.
- Updates will be communicated accordingly.
- The Docker application is **not permitted** for use during assessments.
- Answers supported by the Excel Helper will be accepted. If any dispute arises due to differences in the Excel Helper version used, the case will be reviewed accordingly.

Action Research

- As part of our ongoing effort to enhance your learning experience in C245 Data Analytics with GenAI, we are conducting a research study exploring how explainer videos can support your understanding of key concepts.

- These short explainer videos are designed to recap and reinforce the essential learning points from each lesson, making it easier for you to grasp and retain the <u>**key**</u> concepts covered.

- We will be running two exciting experiments during Week 3 and Week 7. We hope you'll find them a valuable part of your learning journey!

- In both weeks, you'll be invited to complete a short quiz before the explainer video is released. After watching the video, a follow-up quiz will help us understand how much the video has supported your learning. Your participation is greatly appreciated!

- Please note that the quiz scores will **NOT** contribute to your final assessment grade, so there's no pressure at all! That said, we warmly encourage you to give it your best effort, as it's a wonderful opportunity to reinforce and consolidate what you've learned.

- Finally, a short survey will be shared with you afterwards so you can let us know how useful the video has been in your learning. Your feedback is incredibly valuable and will help us continue improving the course for you.

Thank you